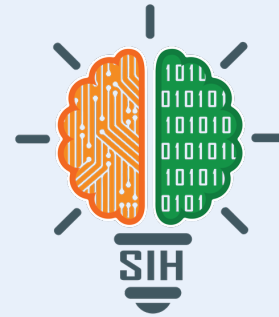


TITLE PAGE

SMART INDIA HACKATHON 2026

Problem Statement ID	SIH26158
Problem Statement Title	Single-Pass Drone Video to Accurate 3D Model Generation System
Theme	Robotics and Drones
PS Category	Software
Organisation	National Technical Research Organisation (NTRO)
Team ID	
Team Name	sixth harmony
Idea Title	OnePass: measurement-grade 3D from a single drone flight



**SMART INDIA
HACKATHON
2026**

PROPOSED SOLUTION (DESCRIBE YOUR IDEA/SOLUTION/PROTOTYPE)

OnePass

Measurement-grade 3D from a single drone flight

In plain terms: fly a drone over an area **once**, and OnePass turns that one video into an accurate, map-ready 3D model. No return trips, no markers placed on the ground first.

WHAT IT DOES

- Builds a 3D model of terrain, buildings, roads and vegetation from one ordinary flight video
- Produces three tiers from that one flight: a fast rough preview, a realistic 3D scene, and a survey-accurate model
- Places the model at its real-world coordinates using the drone's own GPS and altitude sensor

WHY THIS MATTERS

- Standard drone mapping needs several overlapping flights and physical ground markers placed beforehand (ref. 10)
- In a disaster zone or restricted area, teams usually get one flight and no time to place markers

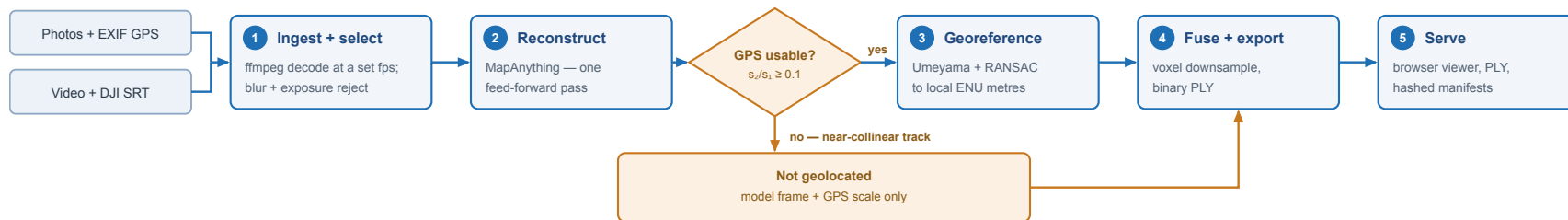
BUILT SO FAR **TIER 1 RUNNING** — photos or video + telemetry → GPS-aligned coloured point cloud in ~30 s, fully offline, with per-point confidence and a browser viewer. Tiers 2–3 and measurement tools are **not** built. Details on slide 3.

A single pass cannot observe every surface. We separate measured geometry from inferred geometry and let the operator see which is which.

TECHNICAL APPROACH

Methodology and process for implementation (Flow Charts/Images/working prototype)

NEURAL TRACK — IMPLEMENTED, RUNS END TO END **BUILT**

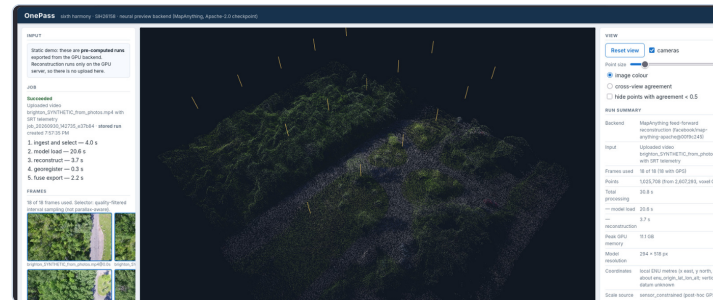


Parameters as implemented: DJI SRT matched by timestamp, ≤ 0.5 s gap, no interpolation · blur reject below 25% of the set's median Laplacian variance · map-anything-apache @ 00f9c245, bf16, GPU capped at 48% · refusal threshold $s_2/s_1 < 0.1$ · binary PLY carrying per-point cross-view agreement, confidence and view id. Modules: selection.py · infer.py · georef.py · fuse.py · server.py.

MEASURED — A100 80GB, 18 aerial photos: **1,134,767 points in 29.8 s** (18.7 s model load, 5.9 s reconstruct), 11.1 GB peak GPU. Agreement with the photos' own GPS 1.43 m RMSE. Offline run verified in a network-less namespace.

PLANNED — NOT IMPLEMENTED

- Parallax-aware keyframe selection (current selector is interval-based)
- Same frames through the COLMAP track; report inter-method agreement
- Tier 2–3: dense MVS / Gaussian splatting, Poisson mesh
- Moving-object masking; LAS/GLB/OBJ/GeoTIFF export; measurement tools



Working prototype — a stored run: video + SRT, 18 frames, 1,025,708 points. Live at bhuvan171.github.io/onepass-neural (ref. 15).

Fully offline and on-premise — no cloud, no external API. **Apache-2.0** MapAnything weights; the CC-BY-NC checkpoint is never downloaded. plyfile (GPL-3.0) and the system Ffmpeg build are still under licence review.

FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

PUBLISHED RESEARCH VS. WHAT WE HAVE MEASURED

What we're measuring, in plain terms	Published research finding (ref. 1)	OnePass: measured so far
How much of the map has no holes, with only a few, barely-overlapping photos of the area	Newer AI methods leave fewer blank patches than the older standard method	Not yet tested. To be confirmed by running the same test on our own system
How far the model's position can be off from the real spot on the ground	Not tested in the cited study: it didn't use GPS	The aligned cloud agrees with the photos' own GPS to about 1.4 m. That is agreement with input GPS, not ground accuracy, which is not yet measured

1. Wu, Landgraf, Ulrich & Qin (2025), UseGeo benchmark, full citation on slide 6.

EVALUATION CRITERIA

Metric	Proves
Single strip vs full grid, same site	The core question
Position error vs known ground points	Real-world accuracy
Size error over known distances	Correct real-world scale
Surface match vs laser scan	Shape accuracy
Coverage, counting only reliable areas	Honest coverage

RISKS AND MITIGATIONS

Risk	Mitigation
Sideways shape is less certain with only one flight path	GPS-constrained alignment; refuses tracks that cannot fix rotation BUILT Adaptive keyframes PLANNED
Moving vehicles and people	Segmentation and motion masking PLANNED
Unseen / occluded surfaces	Per-point confidence shown, low-agreement points can be hidden BUILT

IMPACT AND BENEFITS

Potential impact on the target audience · Benefits of the solution (social, economic, environmental, etc.)

The bottleneck: time

Traditional

Several overlapping flights, ground markers placed beforehand, and processing back at base.

With OnePass

One flight, no markers, and processing on a laptop at the point of capture.

Measured on the prototype: 18 aerial photos became a 1.13 million-point coloured cloud in 29.8 s on one GPU. This is a photo set, not a full-length flight video, so it says nothing yet about long flights.

OPERATIONAL

- Removes the ground-marker survey step (ref. 13)
- Runs on a laptop, offline, at the point of capture

SECTORS

- Disaster and flood response, where teams often get one flight (ref. 12)
- Border mapping, infrastructure, archaeology

ECONOMIC

- No survey crew or RTK-GPS equipment needed on site (ref. 13)
- No repeat flights, no per-seat licence

ENVIRONMENTAL & SOCIAL

- Fewer flight hours per survey
- Faster response than traditional mapping (ref. 11)

11. Overwatch Mapping (2026), drone disaster-response reporting. 12. CRASAR, Texas A&M, Hurricane Harvey UAV deployment (DroneLife, 2025). 13. Industry cost estimates for RTK ground-control survey crews and equipment. Full citations on slide 6.

RESEARCH AND REFERENCES

Details / Links of the reference and research work

1. Wu, Landgraf, Ulrich & Qin (2025). *An Evaluation of DUS3R/MAST3R/VGGT 3D Reconstruction on Photogrammetric Aerial Blocks*. arXiv:2507.14798
2. UseGeo benchmark: github.com/3DOM-FBK/UseGeo
3. UseGeo official evaluation code: github.com/UseGeoEvaluation
4. Florea, Miclea & Nedevschi (2021). *WildUAV: Monocular UAV Dataset for Depth Estimation Tasks*. IEEE ICCP: github.com/hrflir/wuav
5. GauU-Scene / U-Scene: large-scale UAV dataset with LiDAR ground truth
6. OpenDroneMap example datasets: github.com/OpenDroneMap/ODMdata
7. *Effective Strategies for Mitigating the Bowl Effect in UAV Photogrammetry for Corridor Projects*. Drones 9(6):387 (2025)
8. Ferrer-González et al. *UAV Photogrammetry Accuracy Assessment for Corridor Mapping*. Remote Sensing 12(15):2447 (2020)
9. Atik (2025). Evaluation of NeRF and Gaussian Splatting point clouds against photogrammetric reference. Sensors 25(10):2995
10. ScienceDirect (2025). *Optimal ground control point layout for UAV photogrammetry in high-precision 3D mapping*.
11. Overwatch Mapping (2026). *Drone response to storms and disasters: how aerial capture compresses the damage-assessment timeline*.
12. DroneLife (2025). *Drone disaster response project CLARKE*, cites CRASAR/Texas A&M's 119-flight Hurricane Harvey deployment.
13. Aerocartwright (2026). *GCP setup for non-surveyors*, ground-control crew and equipment cost estimates.
14. SkyeBrowse (2026). Drone-to-3D processing time benchmark at scale.
15. **Team repository and measured runs:** github.com/Bhuvan171/onepass-neural
Live viewer: bhuvan171.github.io/onepass-neural
Every figure on slides 3–5 is reproduced in *BUILD_STATUS.md*; permitted claims and their limits in *CLAIMS.md*.